Simple Linear Regression

Regression Analysis

A methodology that studies the relationship between two or more variables.

* 1) Describe a ***relationship*** between two or more variables, e.g. *daily values of two stock indices.*

* 2) ***Predict*** the value of one variable based on the value of another, e.g. *using the amount of training received to predict productivity*

* 3) ***Control*** the level of one variable based on the other, e.g. *Advertisement – Sales*

Observational vs. Experimental Data

* **Observational**

Observe existing historical data. Common in many applications (e.g. stock market data)

- Desirable: it is "**cheap**"
- Undesirable: inference is limited

* **Experimental**

Data is collected via a controlled experiment, common in e.g. marketing studies.

- Desirable: more statistical inference is possible
- Undesirable: could be "**expensive**"

Definitions

* **Response Variable** or **Dependent Variable**

  The variable that we need to predict: Denoted by Y.

* ***Predictor Variable*** or ***Independent Variable***

  The variable we use to predict: Denoted by X

Simple Linear Regression Analysis

* Use to describe a **linear** relationship between one predictor + one response

  e.g. X is the advertisement amount

  — *Predictor Variable*

  Y is sales in a month

  — *Response Variable*

Multiple Linear Regression Analysis

Linear regression with one response and more than one predictor variables

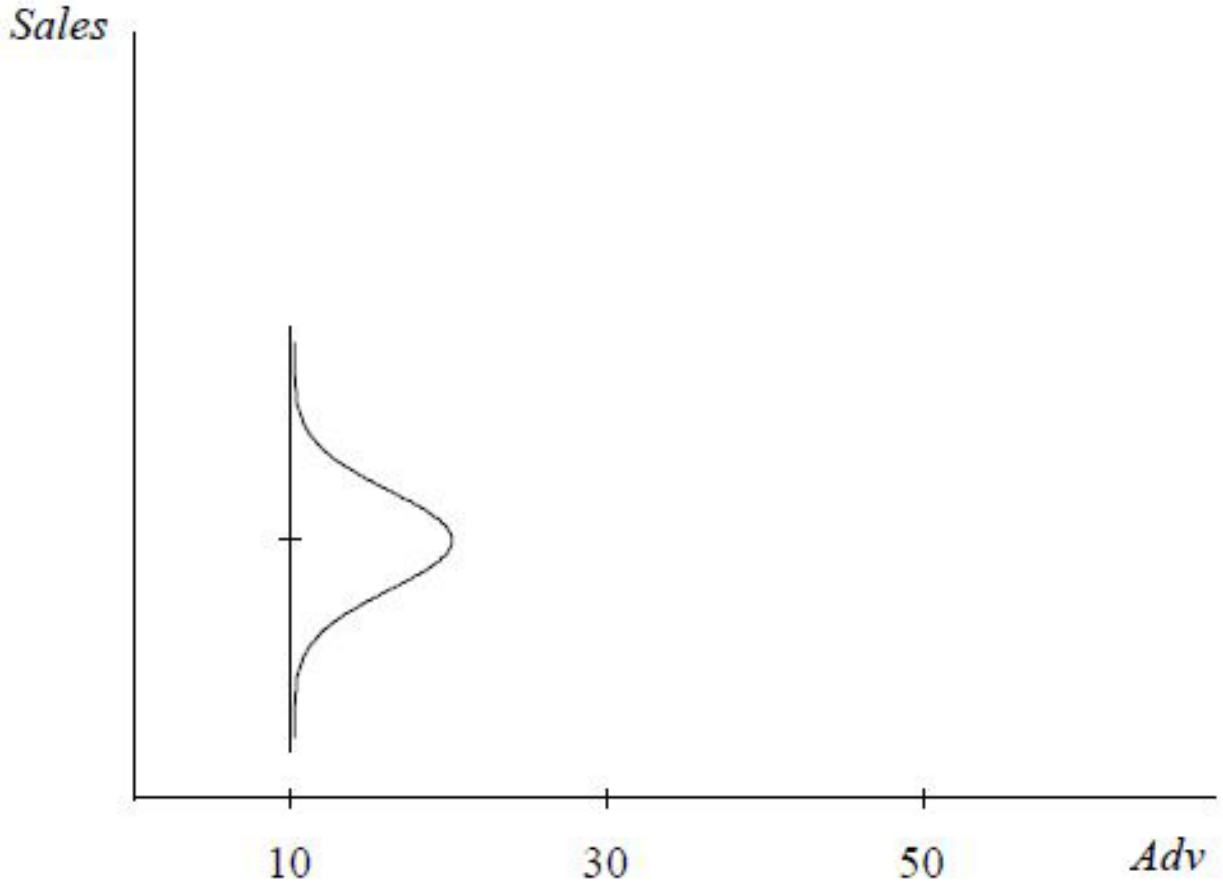
e.g. X_1 *is the advertisement amount*

X_2 *is the price of the product*

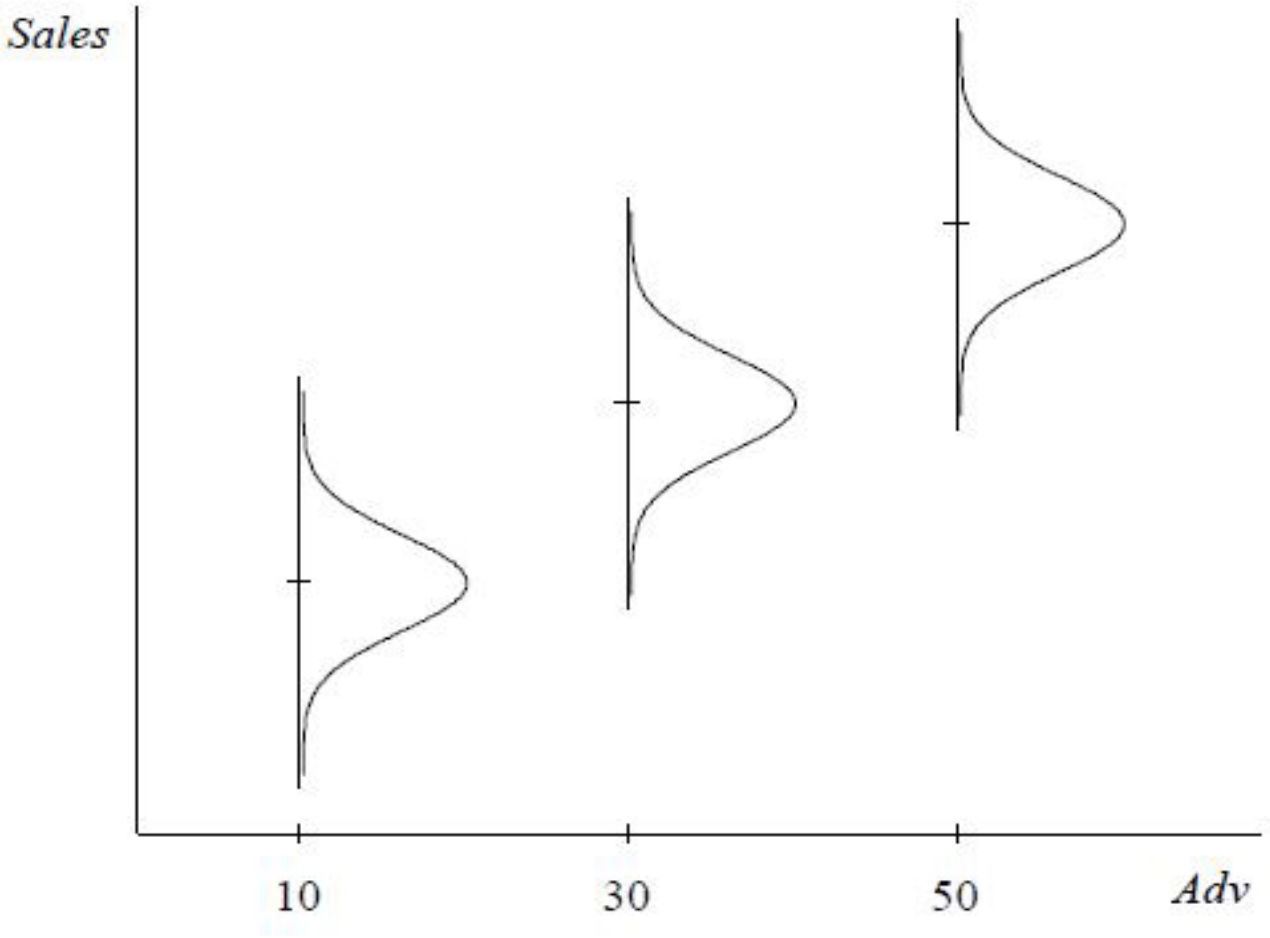
X_3, $X4$, $X5$ …… X_p

Y is sales in a month

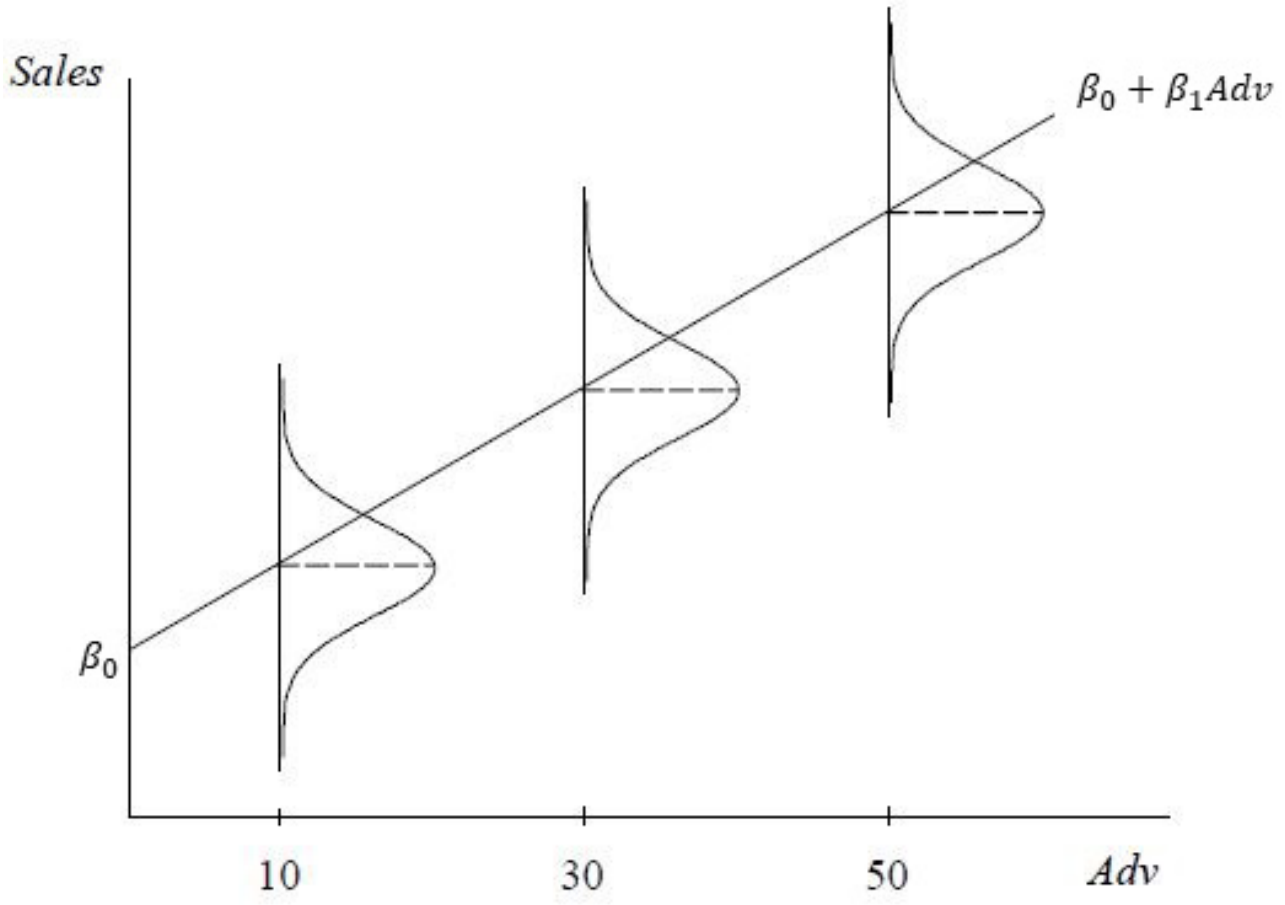
Intuitions for SLRA



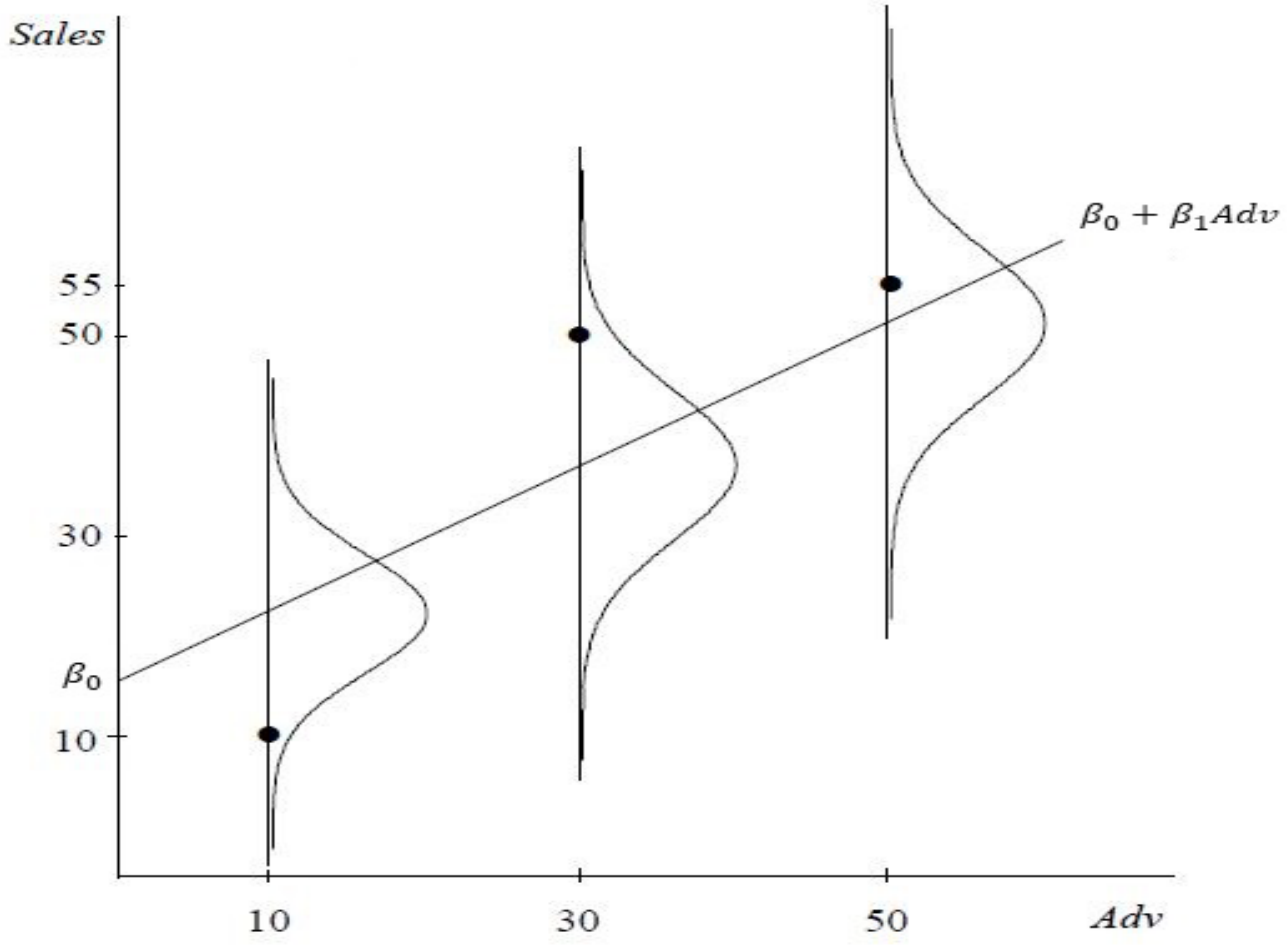
Intuitions for SLRA



Intuitions for SLRA



Intuitions for SLRA



The Simple Linear Regression Model

The Simple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Continue

Y_i

- A particular value of the dependent variable (*this value is* ***random***)

X_i

- A constant that represents a value of the predictor variable (*this value is observed/given and thus, it is* ***constant***)
  X at a particular trial, i

Continue

β_0	• The **intercept**. • The expected value of Y when X = 0
β_1	• The **slope**. • The change in E(Y) for each unit change in X
ε_i	• Denotes the **model error**. • It is random – centered at zero and has constant variance, i.e. $E(\varepsilon_i) = 0, Var(\varepsilon_i) = \sigma^2$

Model Assumptions

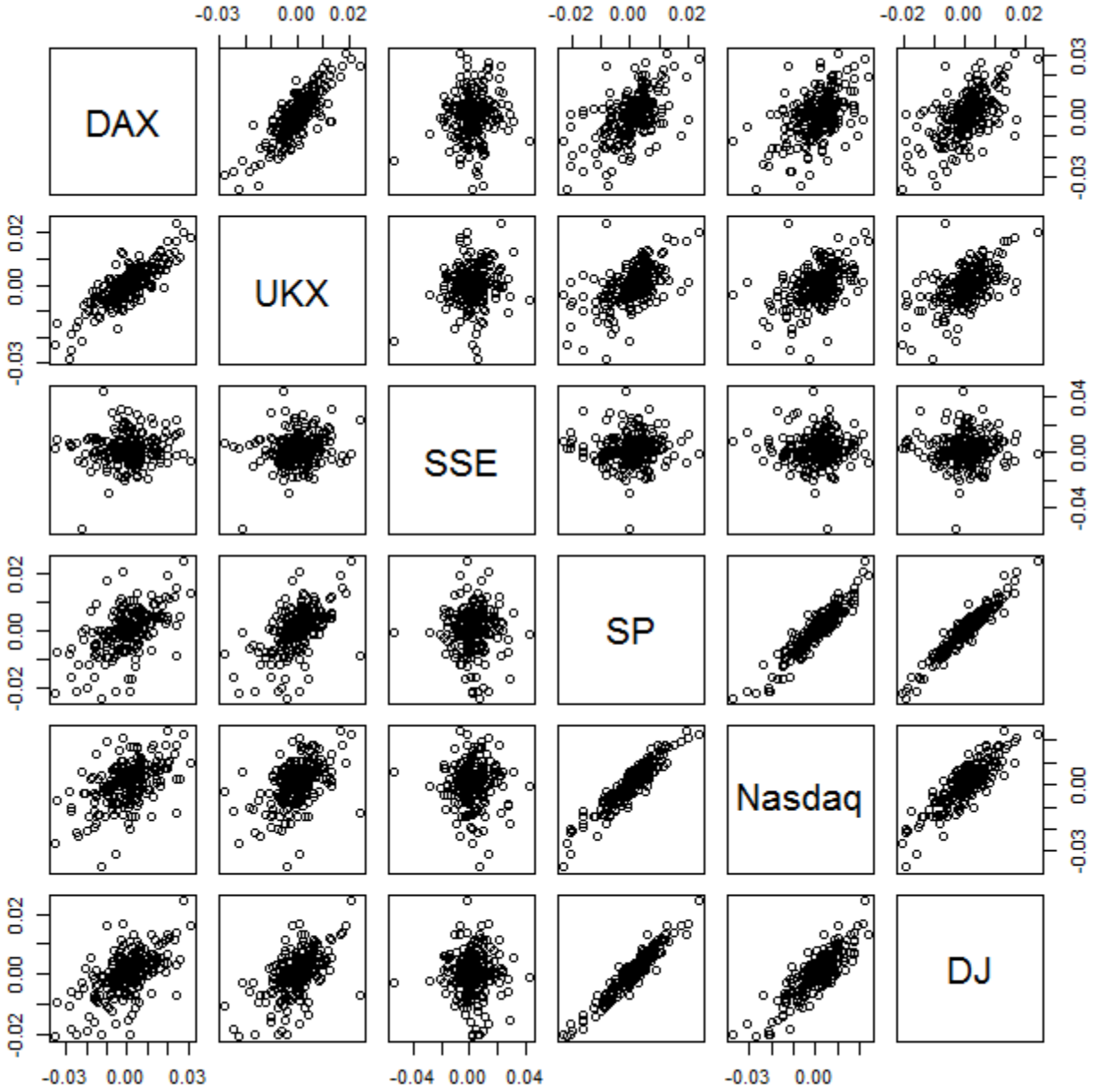
- X_i are given (i.e., not random)
- ε_i are centered at zero, $\mathbf{E}(\varepsilon_i) = 0 \Rightarrow \mathbf{E}(Y_i) = \beta_0 + \beta_1 X_i$
- ε_i have constant variance, $Var(\varepsilon_i) = \sigma^2$
- ε_i and ε_j are uncorrelated

Example: How are different stock indices related?

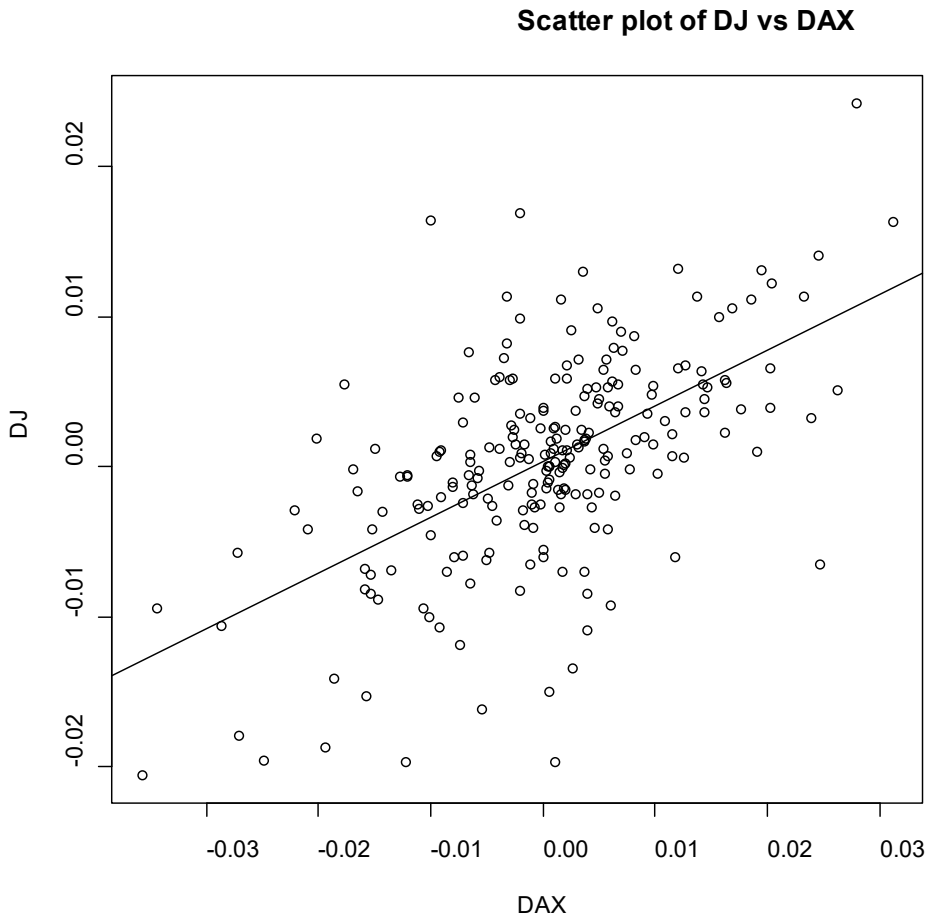


We have daily changes in percentage of the following major stock indices in 2014.

* Dow Jones industrial average
* S&P 500
* Nasdaq
* UKX (a share index of 100 companies listed on the London Stock Exchange)
* DAX (a blue chip stock market index consisting of 30 major German companies traded on the Frankfurt Stock Exchange)
* SSE (a stock market index of all stocks traded at the Shanghai Stock Exchange)


DAX
UKX
SSE
SP
Nasdaq
DJ
-0.03 0.00 0.02
-0.02 0.00 0.02
-0.02 0.00 0.02
-0.03 0.00 0.03
-0.03 0.00 0.02
-0.04 0.00 0.04
-0.02 0.00 0.02
-0.03 0.00
-0.02 0.00 0.02
-0.03 0.00 0.03
-0.04 0.00 0.04
-0.03 0.00

* The last row of the matrix plots gives the scatter plots of DJ (Y axis) vs DAX, UKX, SSE, SP and Nasdaq (X axis)
* Can we use DAX to predict DJ?



Discussion

Let's say we would like to use changes in DAX to predict changes in Dow Jones.

* What is the predictor and what is the response?

* A simple linear regression model appears to be reasonable.

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

* But we have two unknown parameters involved here.

Question

How do we define the line of "Best Fit"?



The Method of Least Squares

* *Define the* ***deviation****:* $Y_i - (\beta_0 + \beta_1 X_i)$
* *Suppose there are n data points, i.e., n pairs of* (X_i, Y_i)
* *Define the* ***sum of squared deviations****:*

$$Q = \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X_i)^2$$

$$= (Y_1 - \beta_0 - \beta_1 X_1)^2 + \cdots + (Y_n - \beta_0 - \beta_1 X_n)^2$$

Continue

Find $\widehat{\beta_0} = b_0$ **and** $\widehat{\beta_1} = b_1$ **such that Q is minimized.**

b_0 and b_1 are the **least squares estimators** for β_0 and β_1

Are b_0 and b_1 random variables?

Notation:

$\overline{X}$: sample mean of $X_i, i = 1, \ldots, n$

$\overline{Y}$: sample mean of $Y_i, i = 1, \ldots, n$

How to find Least Squares Estimates?

$$\frac{dQ}{d\beta_0} = -2\sum[Y_i - \beta_0 - \beta_1 X_i] = 0 \qquad (1)$$

$$\Rightarrow -2\sum(Y_i - b_0 - b_1 X_i) = 0$$

$$\Rightarrow \sum Y_i - nb_0 - b_1 \sum X_i = 0$$

$$\Rightarrow \sum Y_i - b_1 \sum X_i = nb_0$$

$$b_0 = \frac{1}{n}\Sigma Y_i - \frac{b_1}{n}\Sigma x_i$$

$$b_0 = \overline{Y} - b_1 \overline{X}$$

How to find Least Squares Estimates?

$$\frac{dQ}{d\beta_1} = -2\Sigma X_i(y_i - \beta_0 - \beta_1 X_i) = 0 \qquad (2)$$

Replace β_0 and β_1 with their sample estimates

$$\Rightarrow -2\Sigma X_i(Y_i - b_0 - b_1 X_i) = 0$$

Substitute the expression $\overline{Y} - b_1\overline{X}$ for b_0

$$\Rightarrow \Sigma X_i Y_i - \left(\overline{Y} - b_1\overline{X}\right)\Sigma X_i - b_1\Sigma X_i^2 = 0$$

$$\Rightarrow \Sigma X_i Y_i - \overline{Y}\Sigma X_i - b_1\overline{X}\Sigma X_i - b_1\Sigma X_i^2 = 0$$

Since ΣX_i=n$\bar{X}$

$$\Rightarrow \Sigma\, X_i Y_i - \mathrm{n}\bar{X}\bar{Y} - b_1[\Sigma X_1^2 - n\bar{X}^2] = 0$$

$$\Rightarrow b_1 = \frac{\Sigma X_i Y_i - n\bar{X}\bar{Y}}{\Sigma X_i^2 - n\bar{X}^2}$$

Properties of Least Squares Estimator

* b_0 *and* b_1 *are the* **best linear unbiased** *estimator for* β_0 *and* β_1
* **Unbiased** *means* $E(b_1) = \beta_1$*, so on average the estimator hits the target parameter.*
* **Linear** *means* $b_1 = k_1 Y_1 + k_2 Y_2 + \cdots + k_n Y_n$ *for some* $k_1, k_2, \ldots$
* **Best** *means among all possible* ***linear*** **unbiased** *estimators for* β_1*,* b_1 *is the one that has the smallest variance.*
* *The same can be said for* b_0*.*

Properties of k_i

* It can be show that

$$k_i = \frac{X_i - \bar{X}}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

* Properties of k_i
  * $k_1 + \cdots + k_n = 0$
  * $k_1 X_1 + \cdots + k_n X_n = 1$
  * $k_1^2 + \cdots + k_n^2 = \frac{1}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$

Derivation of mean of b_1

$$E[b_1] = E[\Sigma k_i Y_i] = \Sigma E[k_i Y_i] = \Sigma k_i E[Y_i]$$

$$= \Sigma k_i(\beta_0 + \beta_1 X_i) = \Sigma k_i \beta_0 + \Sigma k_i \beta_1 X_i$$

$$= \beta_0 \Sigma k_i + \beta_1 \Sigma k_i X_i$$

$$= 0 + \beta_1 = \beta_1$$

Derivation of variance of b_1

$$var[b_1] = var[\Sigma k_i Y_i] = \Sigma var[k_i Y_i]$$
$$(need\ the\ independence\ of\ the\ Y_i's\ here)$$

$$= \Sigma k_i^2 var[Y_i] = \Sigma k_i^2 \sigma^2 = \sigma^2 \Sigma k_i^2$$
$$= \frac{\sigma^2}{\Sigma (X_i - \bar{X})^2}$$

The Sample Regression Equation

$$\widehat{Y}_i = b_0 + b_1 X_i \rightarrow E(Y_i|X_i) = \beta_0 + \beta_1 X_i$$

b_0

Sample estimate of β_0

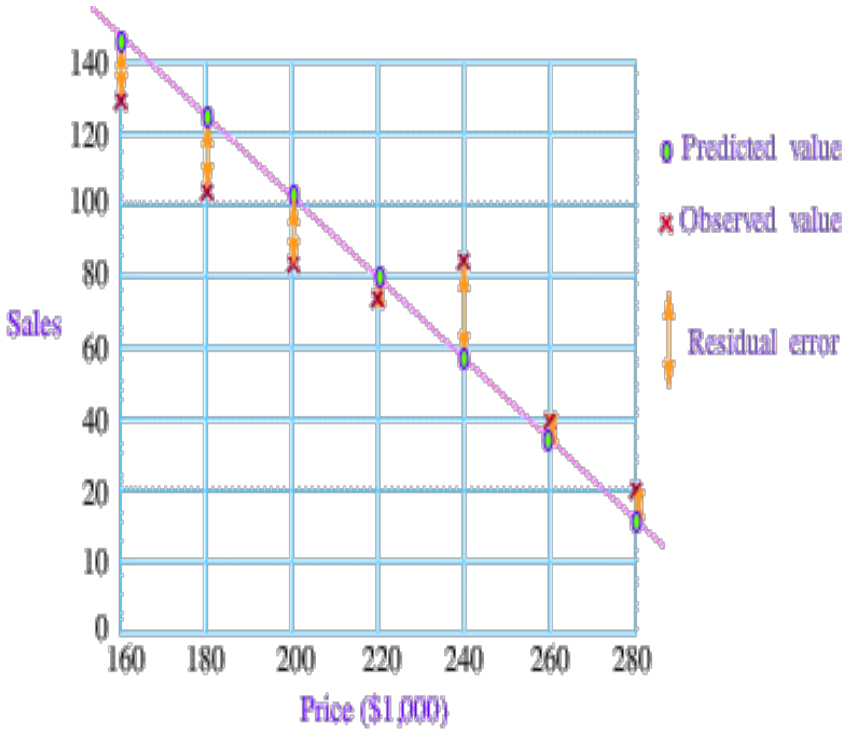
b_1

Sample estimate of β_1

Residual e_i

$$e_i = Y_i - \hat{Y}_i = Y_i - b_0 - b_1 X_i$$

e_i is the residual, the behavior of the residuals gives us insight into the validity of the regression assumptions



Continue

* *Residuals are important for diagnosing whether or not regression assumptions have been met.*
* *Only assumptions so far are* $E(\varepsilon_i) = 0; and\ \varepsilon_i\ and\ \varepsilon_j$ *are uncorrelated.*
* *For least squares (l^2)* ***estimation*** *– these are the only assumptions necessary*
* *For l^2* ***inference,*** *we need more assumptions*

Example: How are different stock indices related?

* What is the **sample regression line**?

* How do you **interpret** the estimated coefficients?

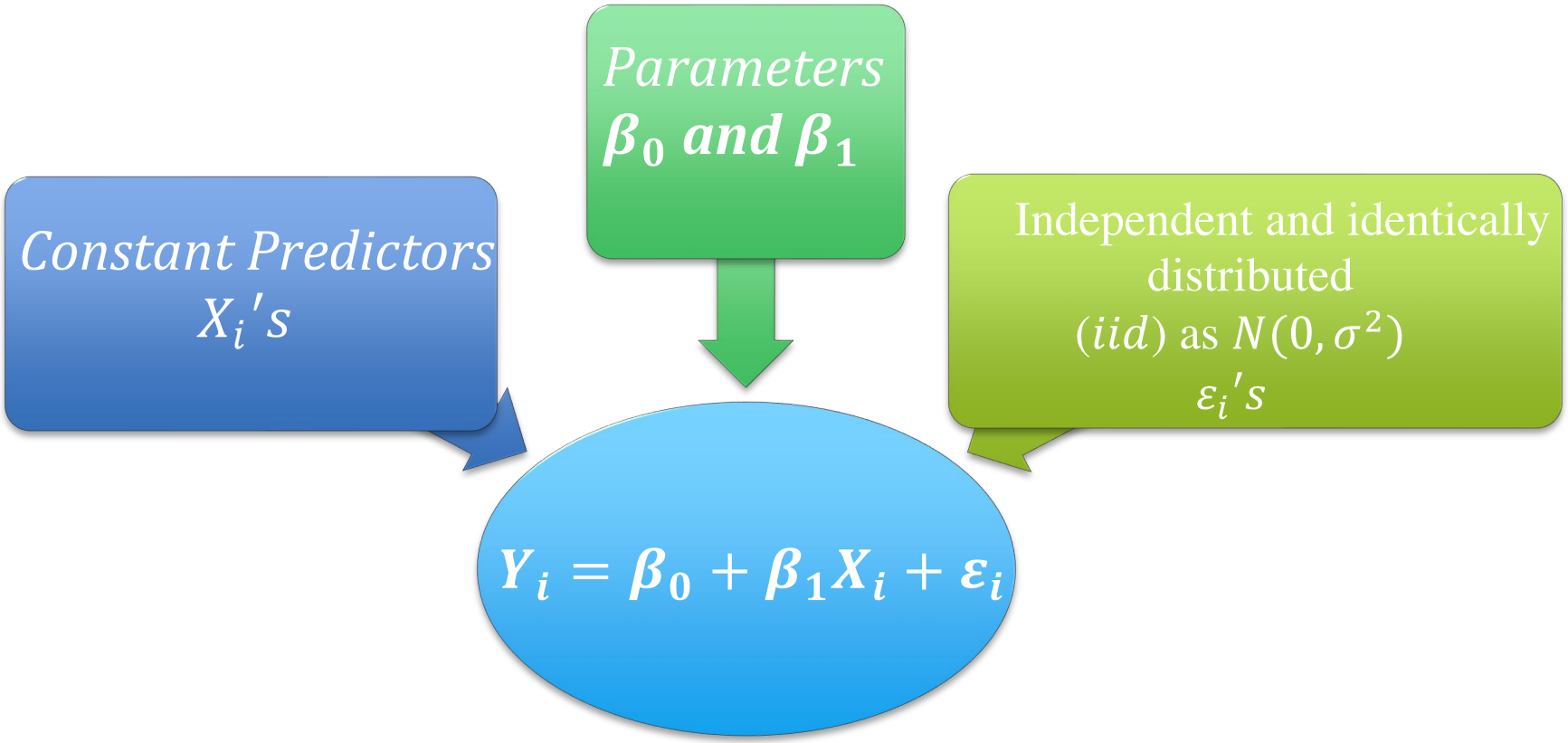


So you have fitted a model, then what?

More questions to ask:

* Does the model help (or do we need to bother)?
* What can we say about the regression coefficients?
* Suppose that DAX change by 1%, how should we **expect** the DJ to change on average? How do we **predict** the actual change of DJ on a **particular** day?
* Does the fitted model fit the data well? Are any model assumptions being violated?
* If there is a problem, what do we do?

A normal error regression model



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* If there is a problem, what do we do?

Some Intuitions

Alg 2	18	33	28	42	49	45	60	48	68	71	93	100

Above are the 2018-2019 Florida Comprehensive Assessment Test (FCAT) scores of algebra 2 for 12 randomly selected students in the Palmetto Middle School.

Now given this information, I am going to randomly select a **new** student who will take the 2019-2020 algebra 2 test . How would you predict his/her score?

Some Intuitions

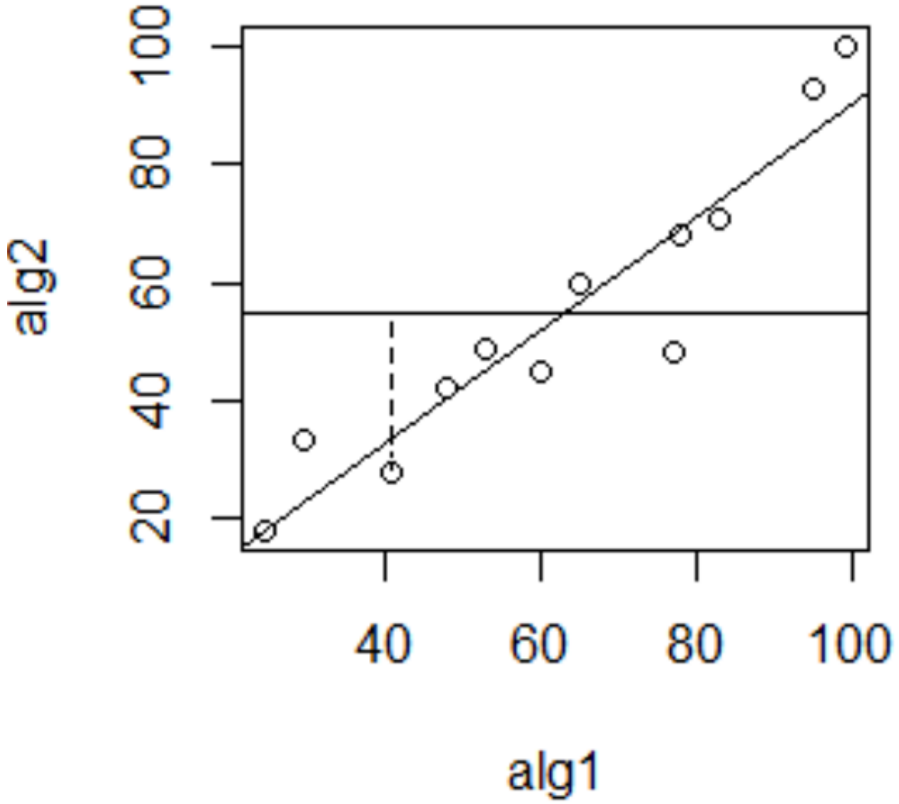
Alg 1	25	30	41	48	53	60	65	77	78	83	95	99
Alg 2	18	33	28	42	49	45	60	48	68	71	93	100

Now suppose the FCAT scores of algebra 1 for the 12 students in the previous year are also known.

For the randomly selected **new** student, his/her algebra 1 test score is also known. How would you estimate his/her new score?

Some Intuitions

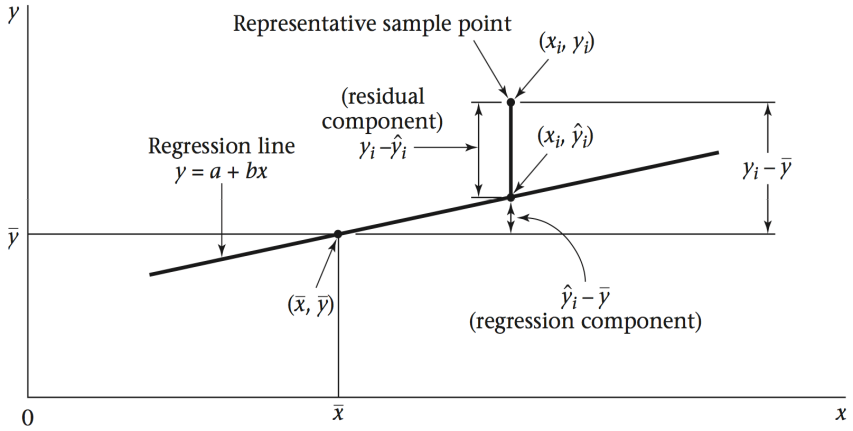
* Now forget about the new student. How would you quantify the performances of the two different approaches in estimating the algebra 2 test scores for **the 12 students in your data**?



Analysis of Variance (ANOVA)

Using ANOVA, we can partition the total variability in Y, i.e., variability around the mean of Y (SSTO) into two parts:

1. the variability about the regression line (SSE)
2. the variability explained by regression (SSR)



Analytically, the partitioning of the SSTO works as follows:

$$SSTO = \sum_{i=1}^{n} (Y_i - \bar{Y})^2$$

$$SSE = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$

$$SSR = \sum_{i=1}^{n} (\hat{Y}_i - \bar{Y})^2$$

SSTO=SSE+SSR.

Partitioning of the degrees of freedom

In statistics, the number of **degrees of freedom** is the number of values in the final calculation of a statistic that are free to vary. (from Wikipedia)

* **Total DF = n-1**
* **DFE = n-2**
* **DFR = 1 = Total DF – DFE**
* **Note that 2 is the number of regression coefficients.**

The ANOVA table

Source	SS	DF	MS	F
Regression	SSR	1	$MSR = \frac{SSR}{1}$	$\frac{MSR}{MSE}$
Error	SSE	n-2	$MSE = \frac{SSE}{n-2}$	
Total	SSTO	n-1		

Estimating σ^2

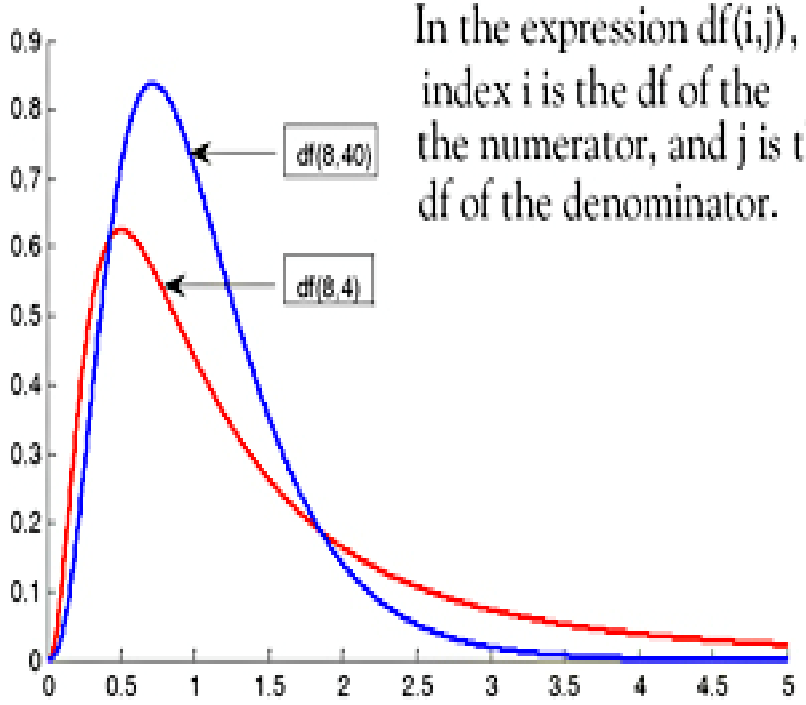
* *Non-regression setup:*

$$S^2 = \sum_{i=1}^{n} (Y_i - \bar{Y})^2 / (n-1)$$

* *Simple Linear Regression:* **MEAN SQUARED ERROR = MSE**

$$= \frac{\sum e_i^2}{n-2} = \frac{\sum (Y_i - \widehat{Y_i})^2}{n-2} = \frac{SSE}{n-2}$$

* The F statistic formed in the above table follows an F-distribution with 1 and n-2 respective degrees of freedom.
* A large value of the F statistic indicates **more variation** explained by the fitted regression model.



F-test

* The F-statistic can be used to test the following hypothesis in simple linear regression:

$$H_0: \beta_1 = 0$$
$$H_a: \beta_1 \neq 0$$

* If $\boldsymbol{H_0}$ is false, what would we expect the F ratio to be?

* The p-value gives us a measure of how far the sample F ratio is from what we expect under the assumption that H_0 is true.

Coefficient of Determination

* A measure of ***linear association***

$$R^2 = \frac{SSR}{SSTO} = 1 - \frac{SSE}{SSTO}$$

$$0 \leq R^2 \leq 1$$

* *R^2 is the proportion of variation in Y that is accounted for by X.*
* *The closer R^2 is to 1, the stronger the **linear** association is between X and Y.*

* If $R^2 = 1$:
  SSR = SSTO; SSE = 0 ⟹ All points lie on regression line.
* If $R^2 = 0$:
  SSE = SSTO; SSR = 0 ⟹ $\overline{Y} = \hat{Y}$
  $\beta_1 = 0$ ⟹ There is **NO** linear association between Y and X.
* *There can be a **strong** relationship even if $R^2 = 0$*

Correlation Coefficient

- *Definition:* $r = \pm\sqrt{R^2}$
- *Sign comes from slope: -1<r<1*
- *The closer |r| is to 1, the stronger the linear association between X and Y.*
- *Similarly, the closer r is to 0, the weaker the* ***linear*** *association.*

Example: How are the different stock indices related?

* Obtain the three sums of squares.

* Obtain the three degrees of freedom.

* Obtain MSR and MSE.

* Obtain the value of the F statistic and verify it using MSR and MSE.

* Explain what the F statistic is used to test for and state your conclusion.

* Obtain the coefficient of determination and correlation coefficient. Explain what they are and what they tell you.